

MATH 101- Mathematical Analysis 1

APPLICATIONS OF DERIVATIVES (PART 3)

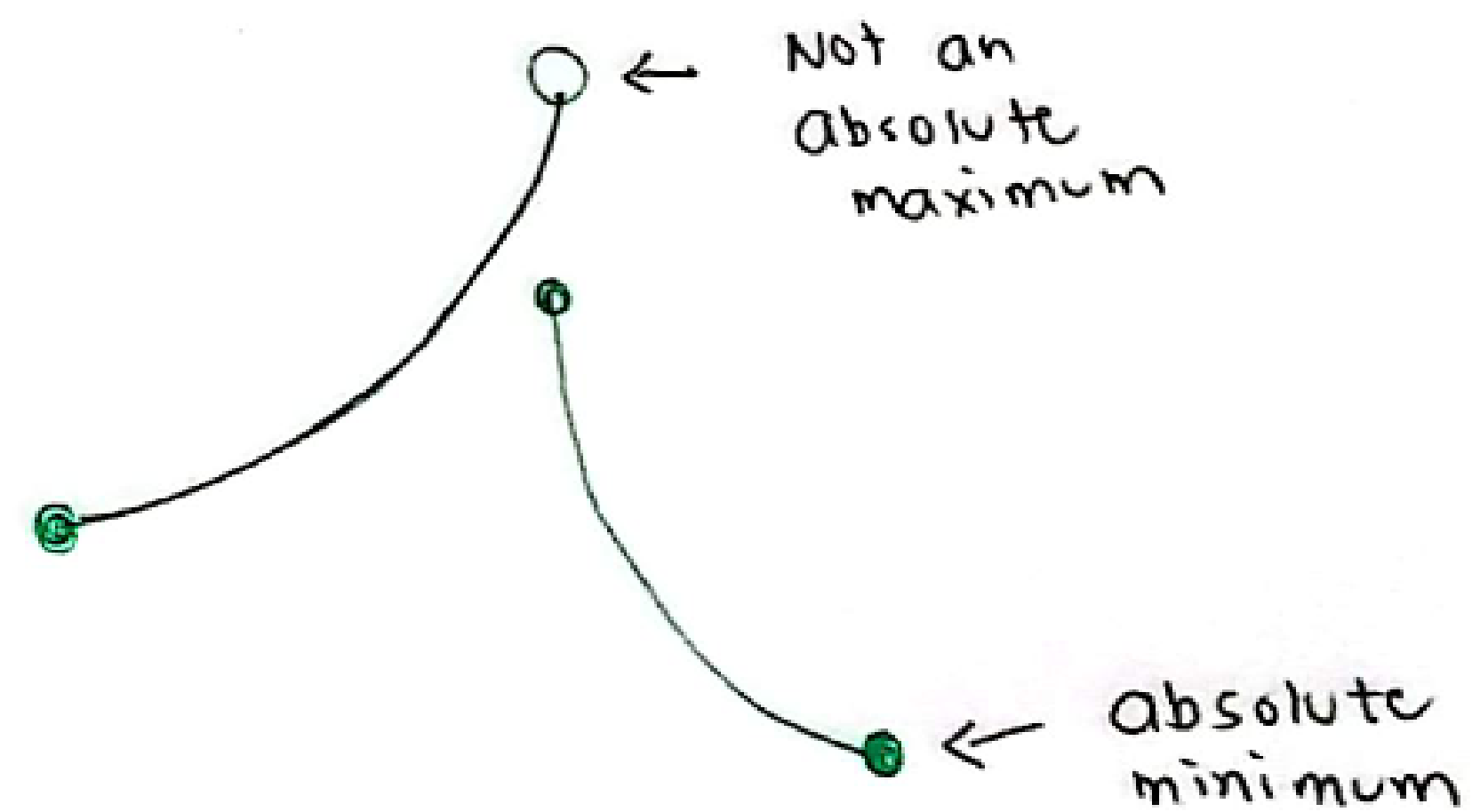
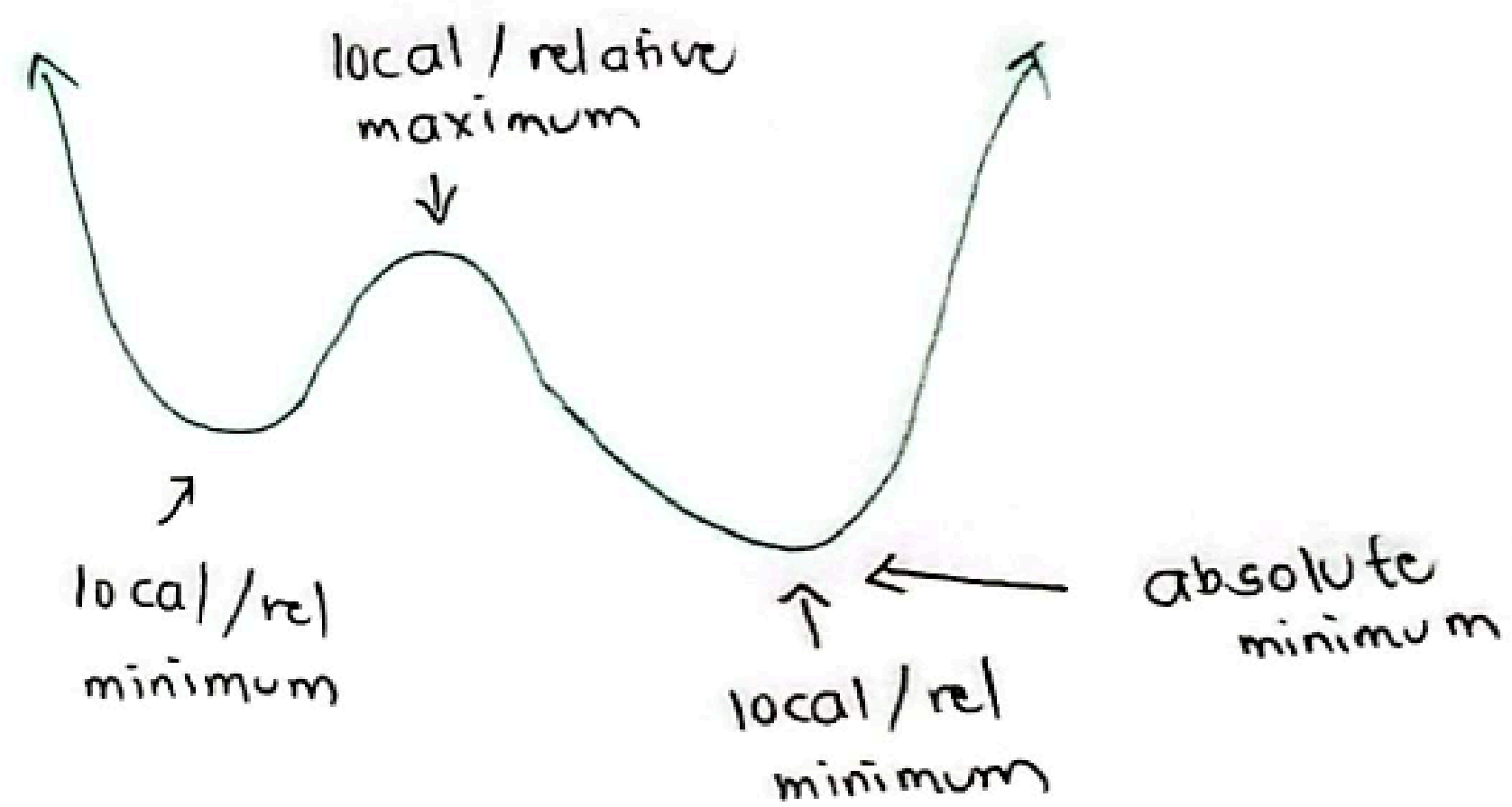
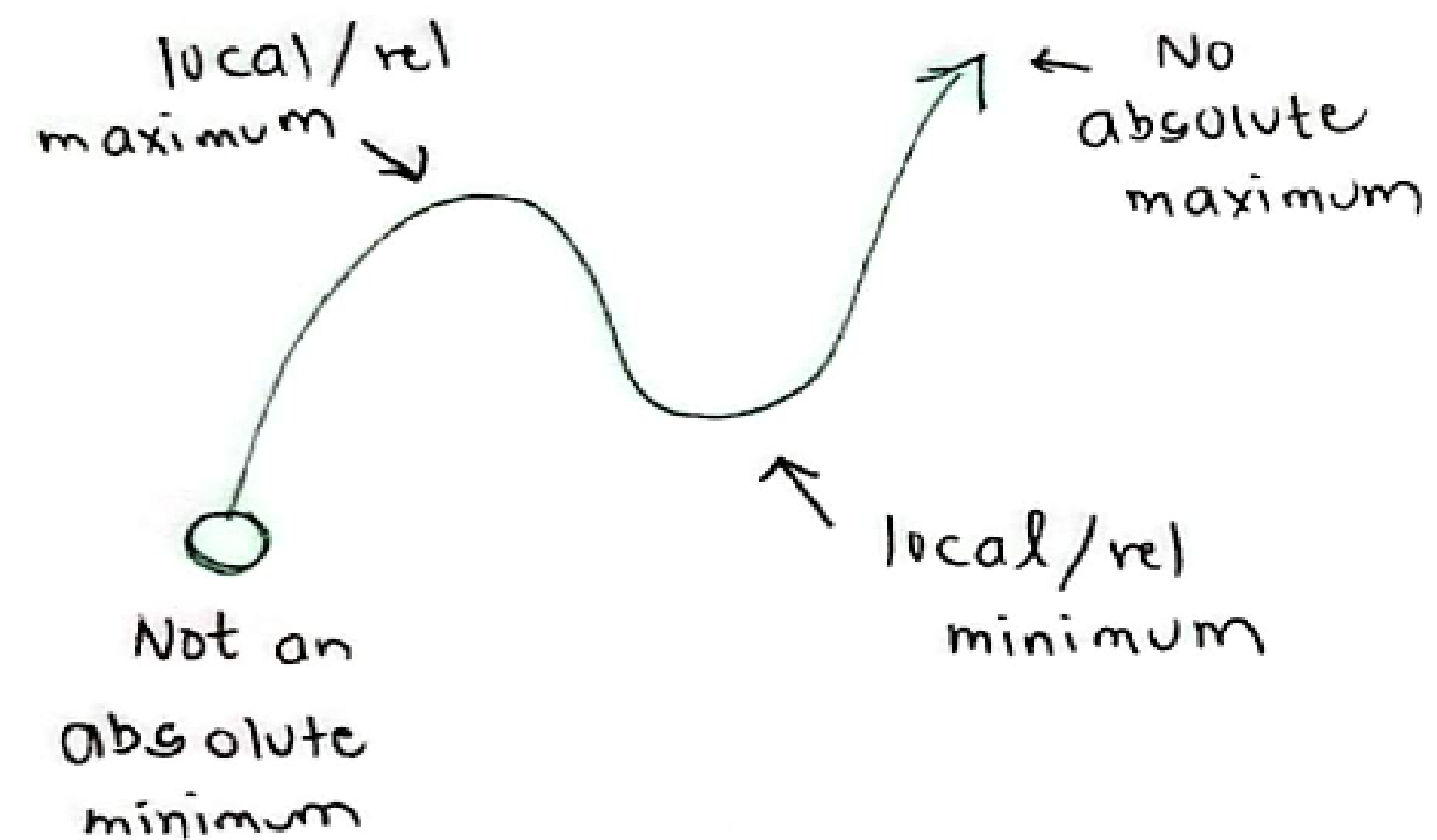
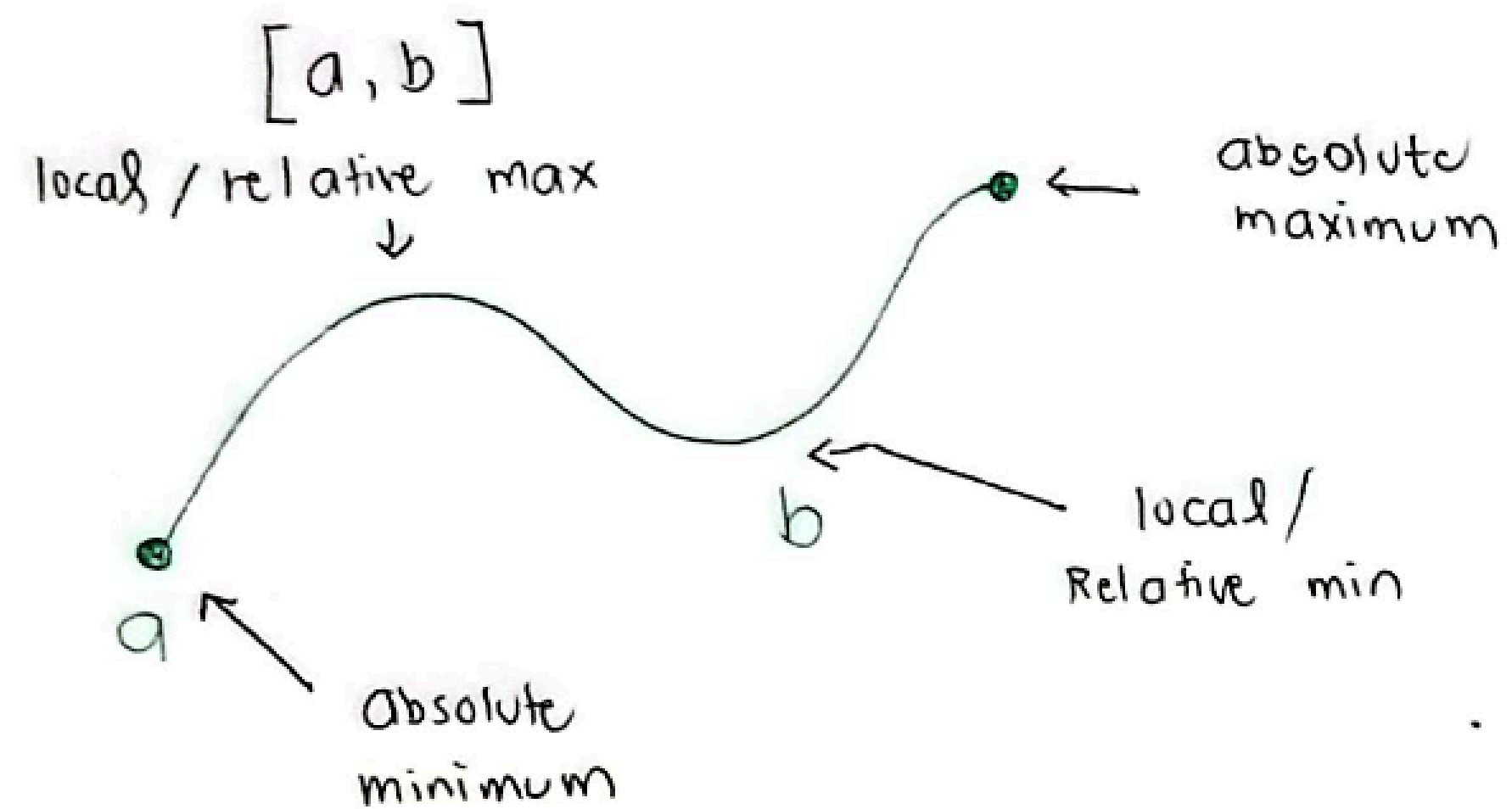
- **Absolute and Relative Extrema**
- **First and Second Derivative Test**
 - **Optimization**

MS. MAYCA N. ODEÑA
Instructor

Absolute maximum and minimum (absolute extrema) are the highest and lowest values a function attains over a given domain.

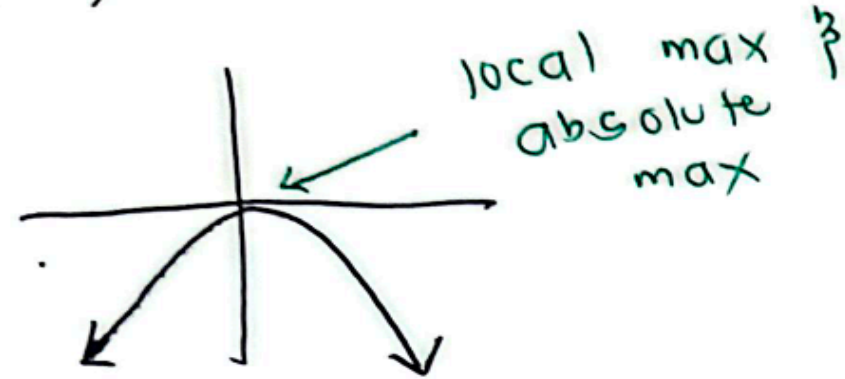
Relative maximums and minimums (also called local extrema) are the peaks (highest) and valleys (lowest) points on a graph in a specific neighborhood, not necessarily the absolute highest or lowest overall. A relative maximum is a peak where the function changes from increasing to decreasing; a relative minimum is a valley where it changes from decreasing to increasing.

A critical number (also called a critical point) of a function f is a value c in the domain of f where the derivative $f'(c)$ is either zero or undefined. These numbers are significant because they are the only places where a function can have a relative maximum or relative minimum.

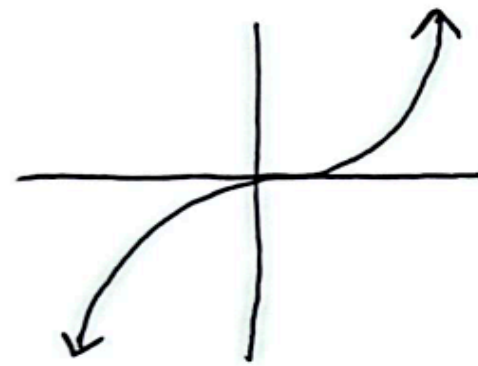


- **EVEN DEGREE** - Has absolute minimum/maximum
- **ODD DEGREE** - No absolute minimum/maximum

$$f(x) = -x^2$$

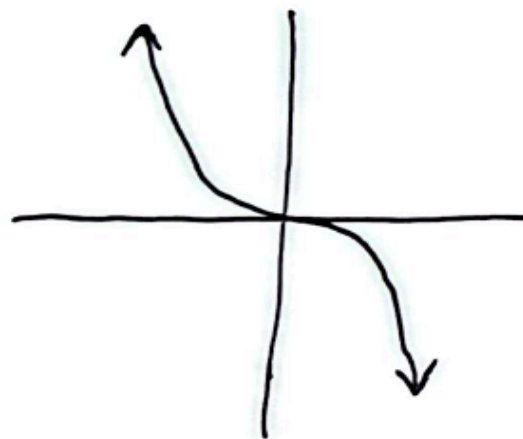


$$f(x) = x^3$$



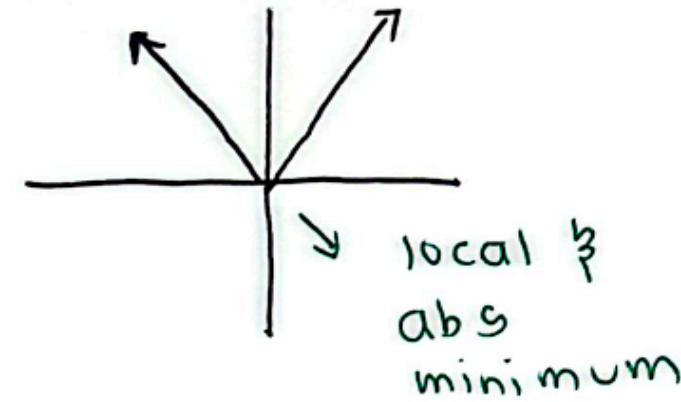
NO absolute
minimum & max

$$f(x) = -x^3$$

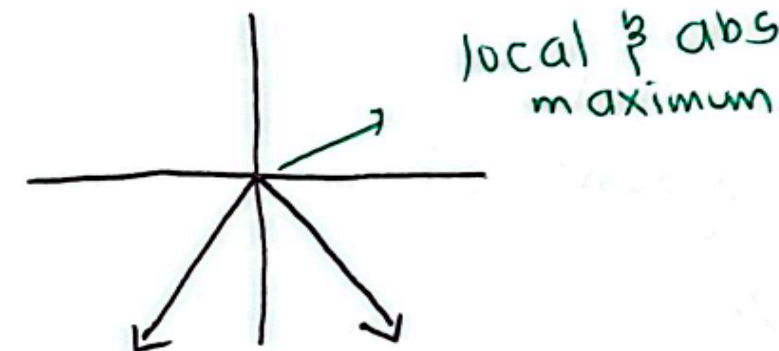


NO abs minimum &
max

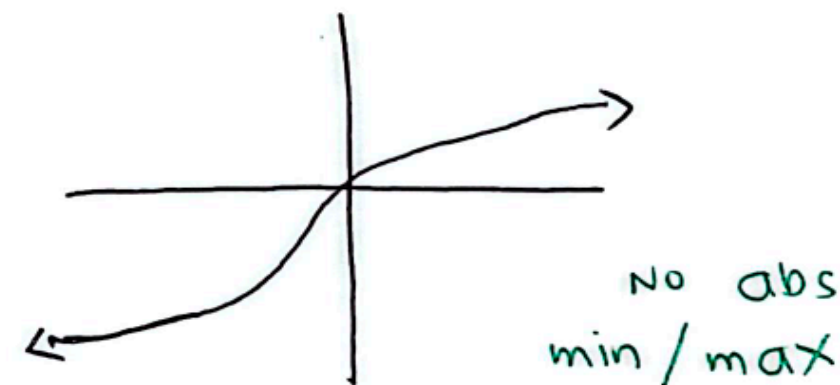
$$f(x) = |x|$$



$$f(x) = -|x|$$

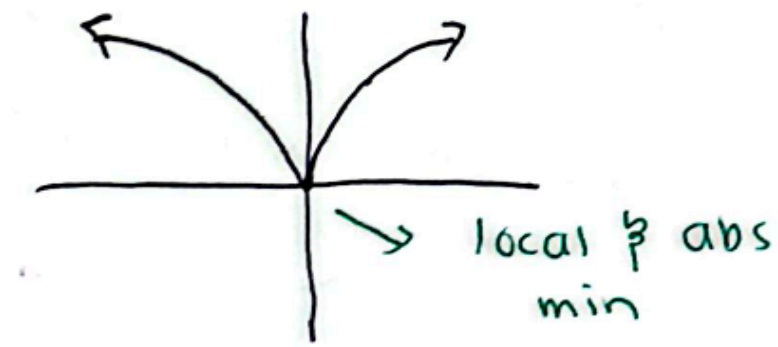


$$f(x) = x^{1/3} = \sqrt[3]{x}$$

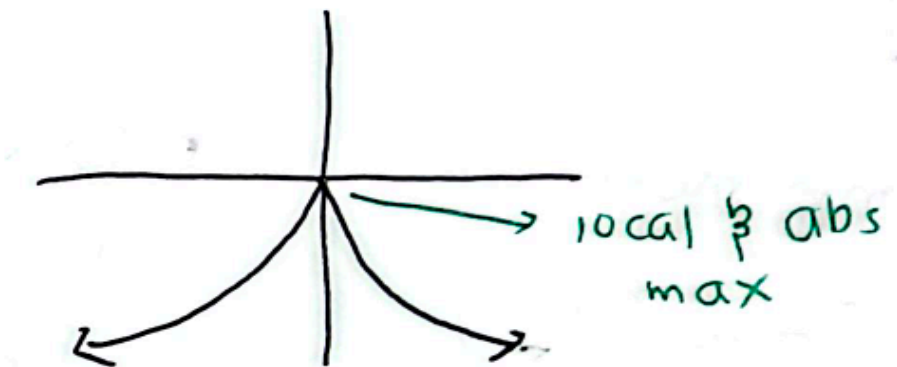


NO abs
min/max

$$f(x) = x^{2/3}$$



$$f(x) = -x^{2/3}$$



FIRST DERIVATIVE TEST

The first derivative test identifies local maxima and minima of a continuous function by analyzing the sign changes of its first derivative, $f'(x)$, around critical points (where $f'(x) = 0$ or is undefined). It determines whether the function is increasing or decreasing, pinpointing where it changes direction.

First Derivative Test Rules

- **Local Maximum:** If $f'(x)$ changes from **positive to negative** at a critical point c , then $f(c)$ is a local maximum.
- **Local Minimum:** If $f'(x)$ changes from **negative to positive** at a critical point c , then $f(c)$ is a local minimum.
- **No Extremum:** If $f'(x)$ does not change sign (e.g., positive to positive), there is no local extremum at c .

Find the critical numbers of the function and identify all extrema.

$$f(x) = 2x^2 - 6x$$

$$f(x) = 2x^2 - 6x$$

SOLUTION:

$$f(x) = 2x^2 - 6x$$

$$f'(x) = 4x - 6$$

$$4x - 6 = 0$$

$$\frac{4x}{4} = \frac{6}{4}$$

$$x = \frac{6}{4} = \boxed{\frac{3}{2}}$$

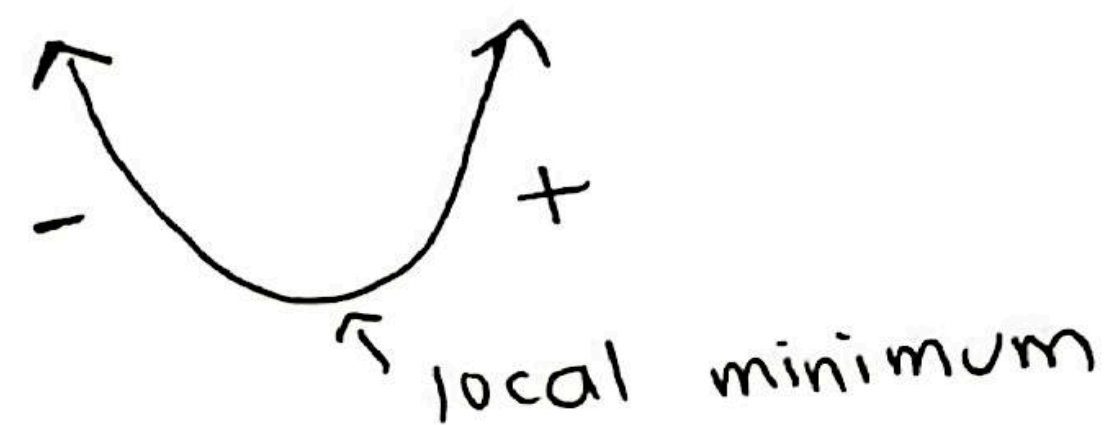
CRITICAL
NUMBER

Value of the minimum:

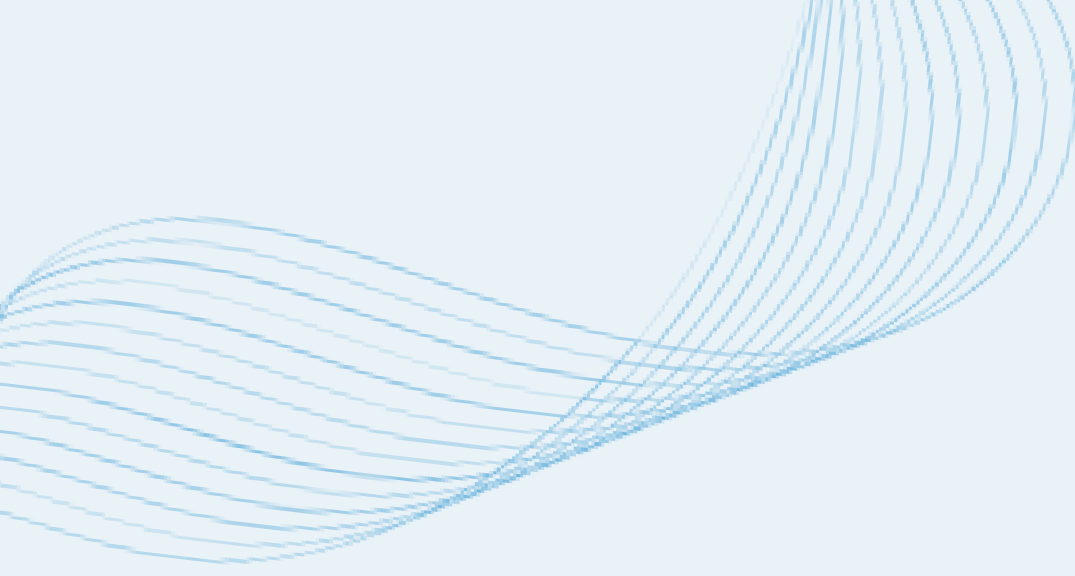
$$\begin{aligned} f\left(\frac{3}{2}\right) &= 2\left(\frac{3}{2}\right)^2 - 6\left(\frac{3}{2}\right) \\ &= \boxed{-\frac{9}{2}} \end{aligned}$$

Parabola opens upward
since $a = 2 > 0$

$\therefore$ minimum at $x = \frac{3}{2}$



Relative / local and
absolute minimum at
 $\left(\frac{3}{2}, -\frac{9}{2}\right)$



Find the critical numbers of the function and identify all extrema.

$$f(x) = 2x^3 + 6x^2 - 48x$$


$$f(x) = 2x^3 + 6x^2 - 48x$$

SOLUTION:

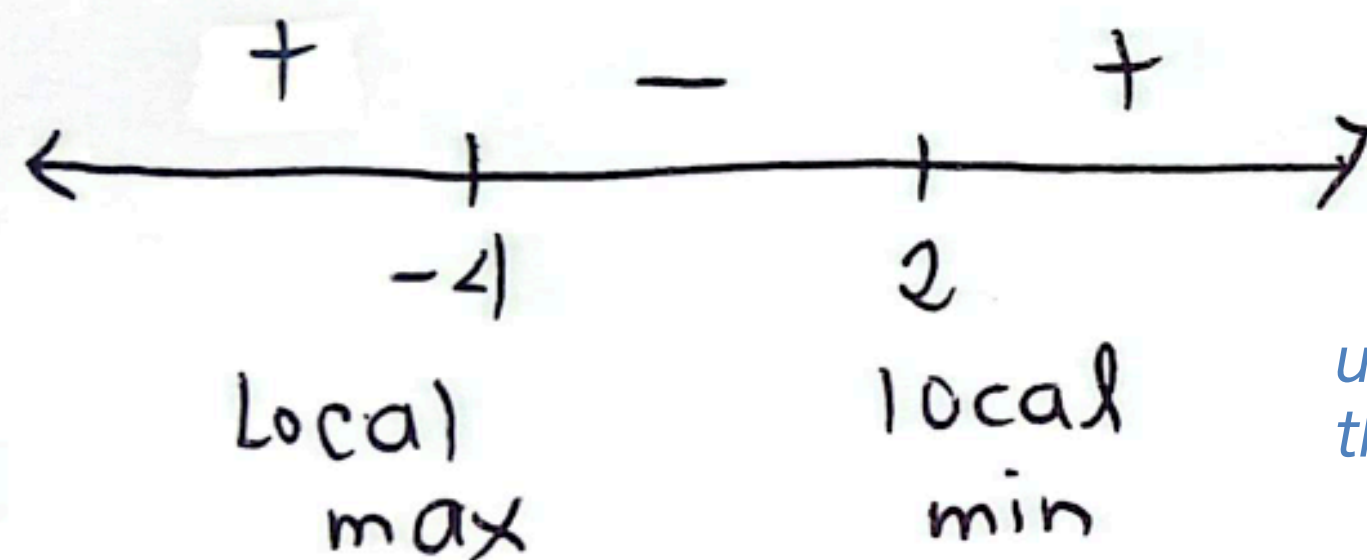
$$f(x) = 2x^3 + 6x^2 - 48x$$

$$f'(x) = 6x^2 + 12x - 48$$

$$6x^2 + 12x - 48 = 0$$

$$6(x+4)(x-2) = 0$$

$$\boxed{x = -4} \quad \boxed{x = 2} \rightarrow \text{CRITICAL NUMBERS}$$



use the first derivative for the sign chart

$$f(-4) = 2(-4)^3 + 6(-4)^2 - 48(-4) = \boxed{160} \rightarrow \text{max}$$

$$f(2) = 2(2)^3 + 6(2)^2 - 48(2) = \boxed{-56} \rightarrow \text{min}$$

Relative / local Max: 160 at $x = -4$

Relative / local Min: -56 at $x = 2$

NO Absolute min / max

Find the critical numbers of the function and identify all extrema.

$$f(x) = 5x^{2/3} + x^{5/3}$$

$$f(x) = 5x^{2/3} + x^{5/3}$$

SOLUTION:

$$f(x) = 5x^{2/3} + x^{5/3}$$

$$f'(x) = 5 \cdot \frac{2}{3} x^{-1/3} + \frac{5}{3} x^{2/3}$$

$$= \frac{10}{3x^{1/3}} + \frac{5x^{2/3}}{3}$$

$$= \frac{10 + 5x}{3x^{1/3}} = \frac{5(x+2)}{3x^{1/3}}$$

$$5(x+2) = 0$$

$$\boxed{x = -2}$$

$$3x^{1/3} = 0$$

$$x^{1/3} = 0$$

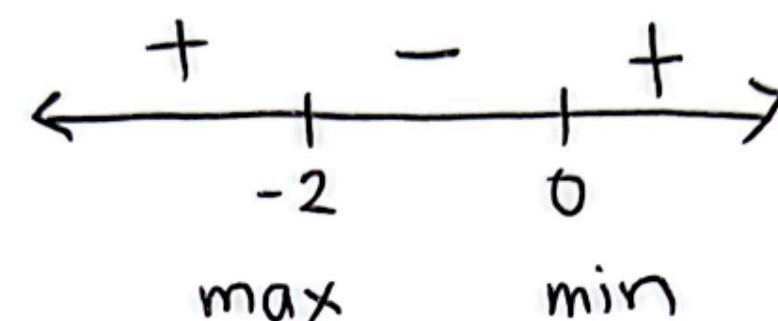
$$\boxed{x = 0}$$

CRITICAL NUMBERS:

$$x = -2$$

$$x = 0$$

↳ continuous
but not
differentiable



Relative/local max: $3\sqrt[3]{4} \approx 4.76$ at $x = -2$

$$5(-2)^{2/3} + (-2)^{5/3}$$

$$= 5\sqrt[3]{4} - \sqrt[3]{32}$$

$$= 5\sqrt[3]{4} - 3\sqrt[3]{4}$$

$$= 3\sqrt[3]{4} \approx 4.76$$

Relative/local min: 0 at $x = 0$

$$5(0)^{2/3} + (0)^{5/3} = 0$$

No Absolute min/max

Find the absolute and relative values of the function:

$$f(x) = 3x^2 - 12x + 5 \text{ at } [0, 3]$$

$$f(x) = 3x^2 - 12x + 5 \text{ at } [0, 3]$$

SOLUTION:

$$f(x) = 3x^2 - 12x + 5 \text{ at } [0, 3]$$

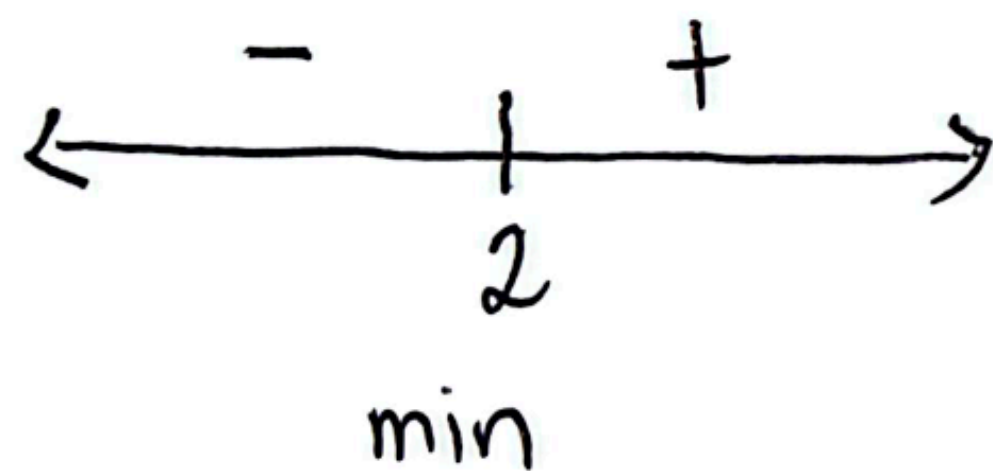
$$f'(x) = 6x - 12$$

$$6x - 12 = 0$$

$$\frac{6x}{6} = \frac{12}{6}$$

$$\boxed{x = 2}$$

CRITICAL NUM



x	y
0	5
2	-7
3	-4

$$f(0) = 3(0)^2 - 12(0) + 5 = 5$$

$$f(2) = 3(2)^2 - 12(2) + 5 = -7$$

$$f(3) = 3(3)^2 - 12(3) + 5 = -4$$

$$\underline{x = 0, y = 5} \rightarrow \text{abs min}$$

$$\underline{x = 2, y = -7} \rightarrow \text{local min \& abs min}$$

NO RELATIVE MAX

Find the absolute and relative values of the function:

$$f(x) = 2x^4 - 4x^2 - 1 \text{ at } [-2, 3]$$

$$f(x) = 2x^4 - 4x^2 - 1 \text{ at } [-2, 3]$$

SOLUTION:

$$f(x) = 2x^4 - 4x^2 - 1 \text{ at } [-2, 3]$$

$$f'(x) = 8x^3 - 8x$$

$$8x^3 - 8x = 0$$

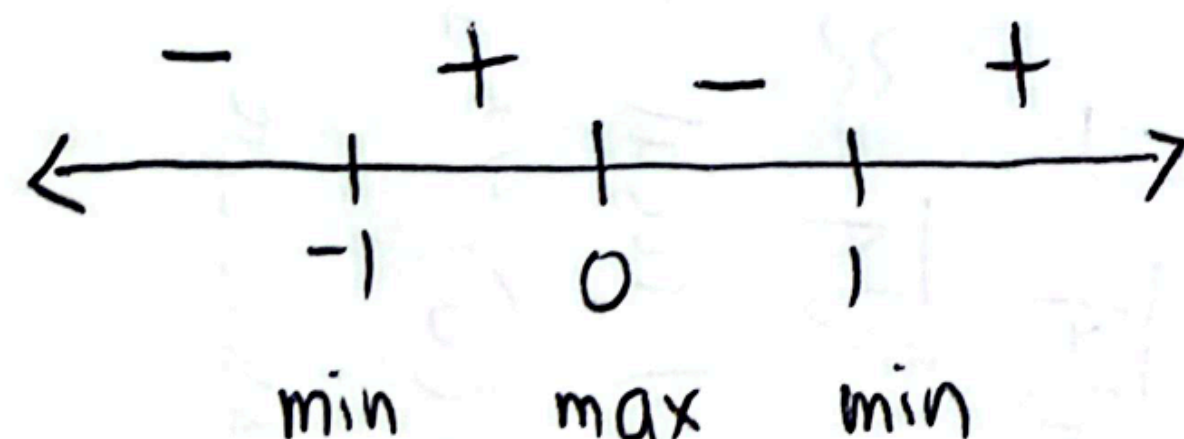
$$8x(x^2 - 1) = 0$$

$$8x(x+1)(x-1) = 0$$

$$8x = 0 \quad | \quad x+1 = 0 \quad | \quad x-1 = 0$$

$$\boxed{x = 0} \quad | \quad \boxed{x = -1} \quad | \quad \boxed{x = 1}$$

CRITICAL NUM.



X	y
-2	15
-1	-3
0	-1
1	-3
3	125

→ Relative min / abs min

→ Relative max

→ Relative min / abs min

→ abs max

Abs max: 125 at $x = 3$

Abs min: -3 at $x = -1, 1$

Rel max: -1 at $x = 0$

Rel min: -3 at $x = -1, 1$

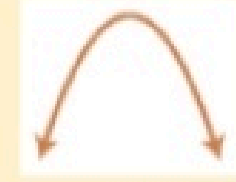
SECOND DERIVATIVE TEST

The second derivative test uses a function's second derivative, $f''(x)$, to determine if a critical point ($f'(c) = 0$) is a local maximum ($f''(c) < 0$, concave down) or local minimum ($f''(c) > 0$, concave up). If $f''(c) = 0$, the test is inconclusive, and the first derivative test must be used.

Second Derivative Test

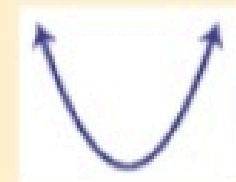
Local Max

$$f'(c) = 0 \text{ and } f''(c) < 0$$



Local Min

$$f'(c) = 0 \text{ and } f''(c) > 0$$



Key Principles and Steps

- **Find Critical Points:** Calculate $f'(x)$, set it to 0, and solve for x to find critical values (stationary points).
- **Evaluate Concavity:** Calculate $f''(x)$.
- **Apply Test:**
 - If $f'(c) = 0$ and $f''(c) < 0$, then $f(x)$ has a local maximum at c .
 - If $f'(c) = 0$ and $f''(c) > 0$, then $f(x)$ has a local minimum at c .
 - If $f'(c) = 0$ and $f''(c) = 0$, the test is inconclusive (could be a minimum, maximum, or inflection point).

Use the second derivative test to determine any relative extrema

$$f(x) = 2x^3 + 6x^2 - 48x$$

$$f(x) = 2x^3 + 6x^2 - 48x$$

SOLUTION:

$$f(x) = 2x^3 + 6x^2 - 48x$$

$$f'(x) = 6x^2 + 12x - 48$$

$$6x^2 + 12x - 48 = 0$$

$$(x+4)(x-2) = 0$$

$$\boxed{x = -4} \quad \boxed{x = 2}$$

CRITICAL NUM

$$f''(x) = 12x + 12$$

$$f''(-4) = 12(-4) + 12 = \underline{-36} < 0$$

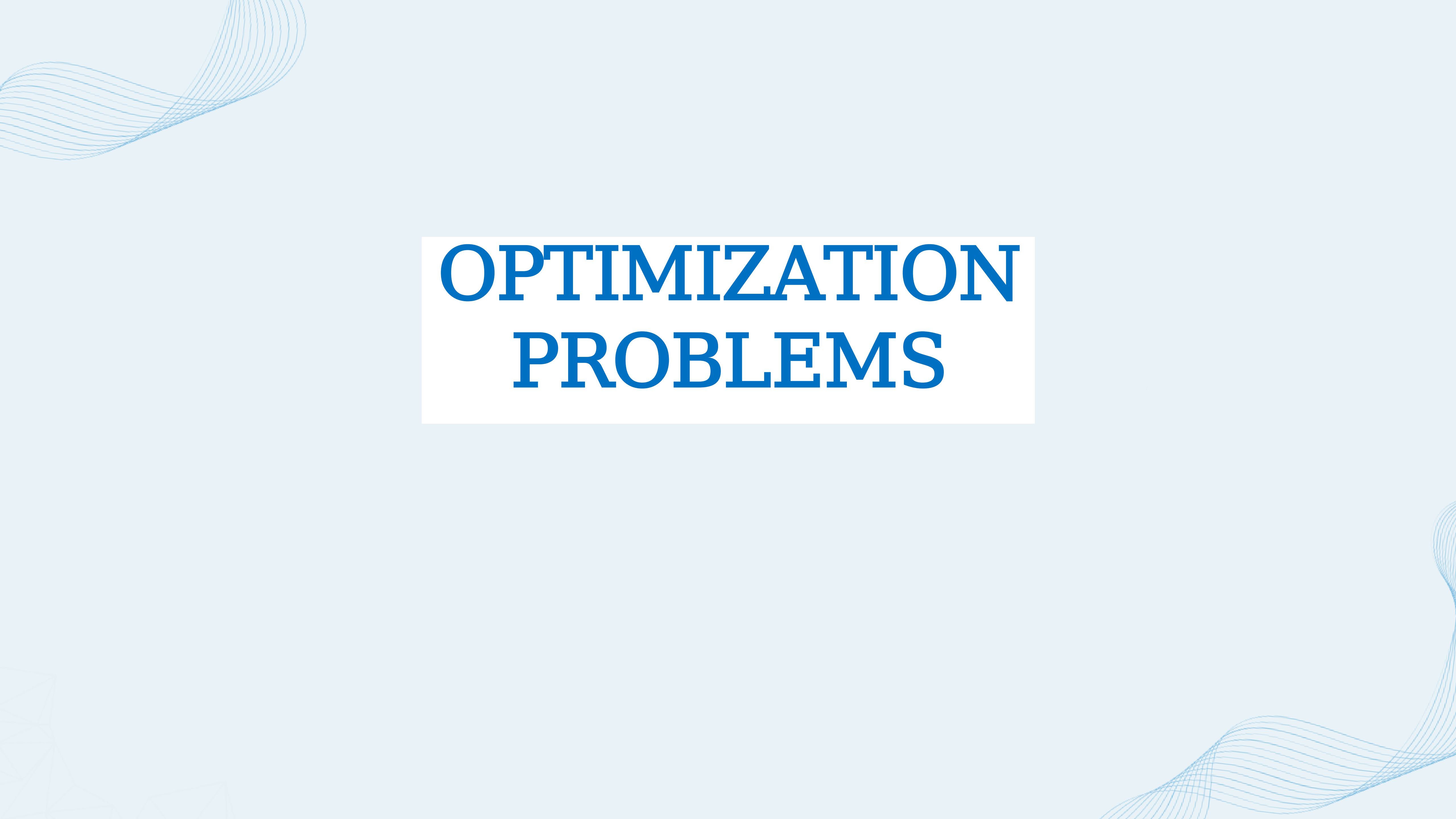
Relative max

$$f''(2) = 12(2) + 12 = \underline{36} > 0$$

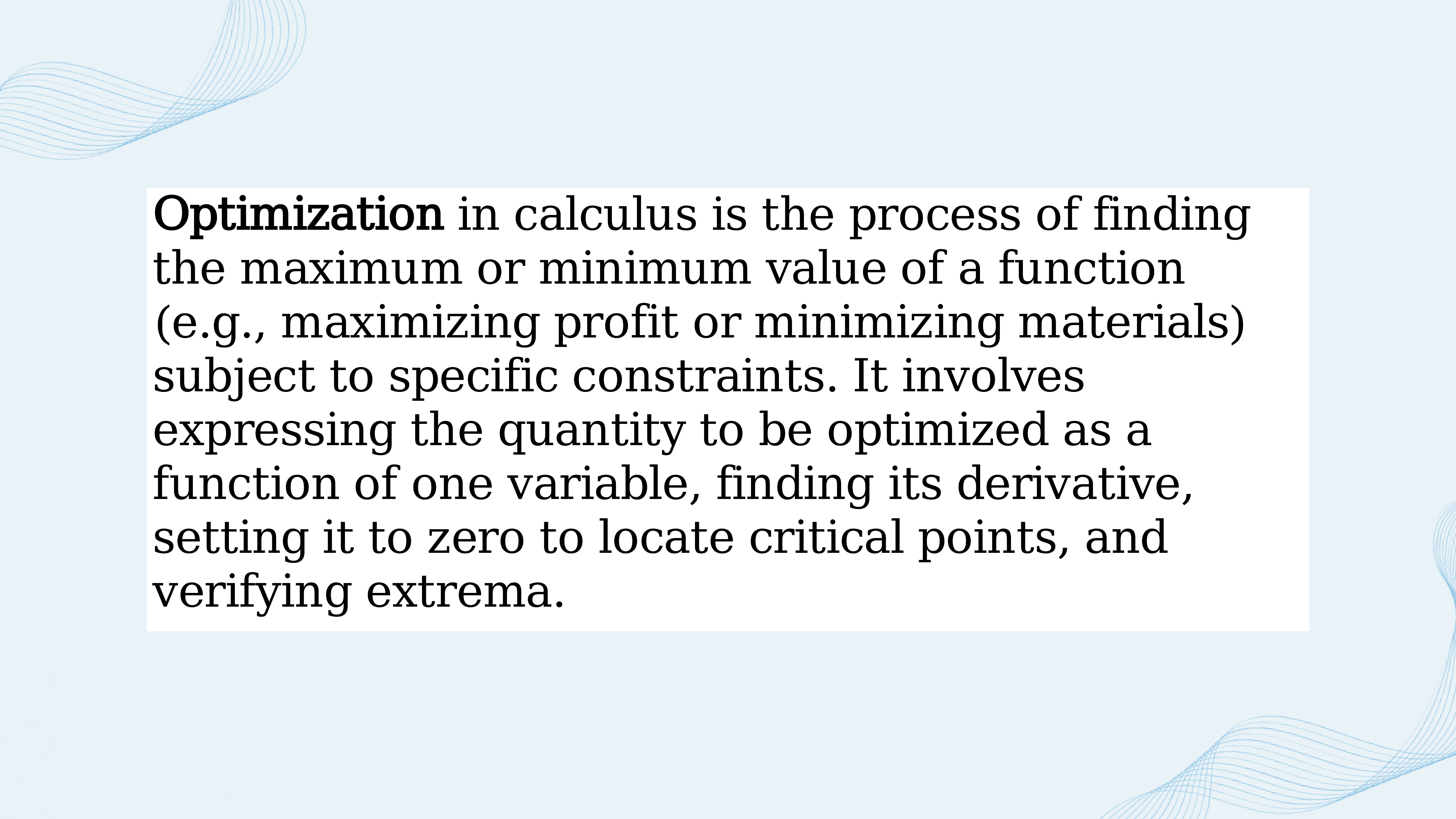
Relative min

$$\begin{aligned} \text{Relative max: } f(-4) &= 2(-4)^3 + 6(-4)^2 - 48(-4) \\ &= \underline{160} \end{aligned}$$

$$\begin{aligned} \text{Relative min: } f(2) &= 2(2)^3 + 6(2)^2 - 48(2) \\ &= \underline{-56} \end{aligned}$$



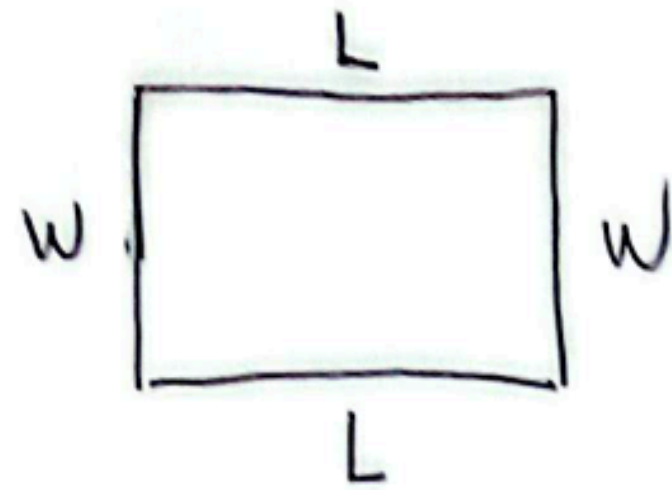
OPTIMIZATION PROBLEMS



Optimization in calculus is the process of finding the maximum or minimum value of a function (e.g., maximizing profit or minimizing materials) subject to specific constraints. It involves expressing the quantity to be optimized as a function of one variable, finding its derivative, setting it to zero to locate critical points, and verifying extrema.

Find the dimensions of a rectangle with a perimeter of 200ft with an area as large as possible.

SOLUTION:



$$P = 2L + 2w$$
$$\frac{200}{2} = \frac{2L + 2w}{2}$$

$$100 = L + w$$

$$\underline{100 - L = w}$$

$$100 - 50 = w$$

$$\boxed{50 = w}$$

$$A = Lw$$

$$A = L(100 - L)$$

$$A = 100L - L^2$$

$$A' = 100 - 2L$$

$$100 - 2L = 0$$

$$\frac{100}{2} = \frac{2L}{2}$$

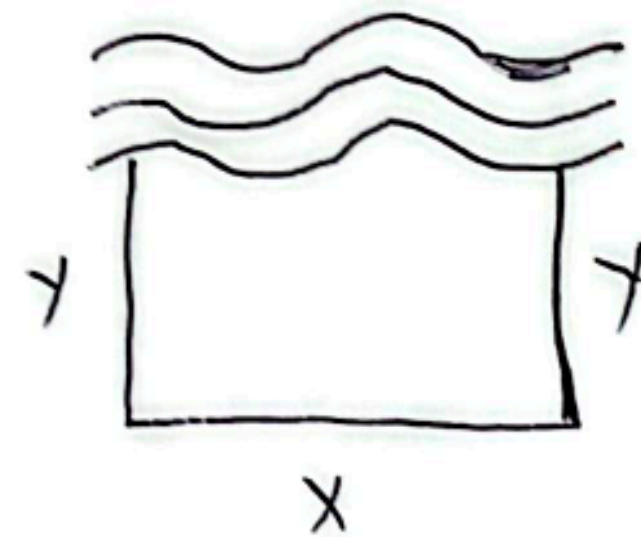
$$\boxed{50 = L}$$

The dimensions are
50 ft by 50 ft.

$$A = (50 \text{ ft})(50 \text{ ft})$$
$$= \underline{2500 \text{ ft}^2}$$

A farmer has 600ft of fencing and wants to create a rectangular field along a river. He needs no fence along the river itself. What are the dimensions of the field that has the largest area?

SOLUTION:



$$\begin{aligned}
 P &= x + 2y \\
 600 &= x + 2y \\
 600 - 2y &= x \\
 \hline
 600 - 2(150) &= x \\
 600 - 300 &= x \\
 \boxed{300} &= x
 \end{aligned}$$

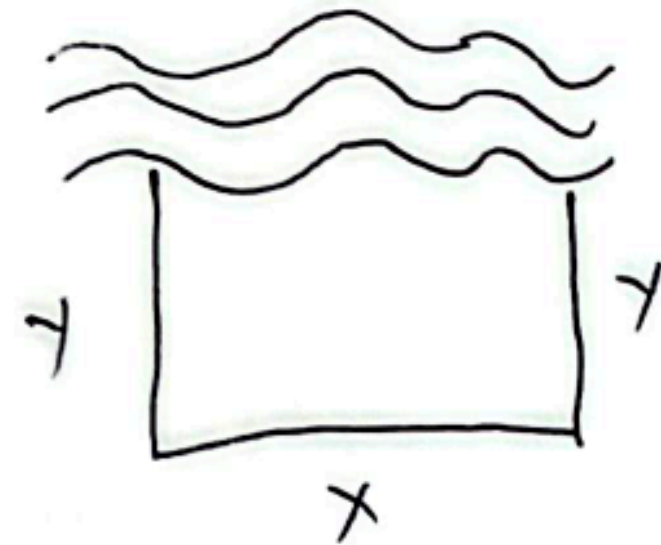
$$\begin{aligned}
 A &= xy \\
 A &= (600 - 2y)y \\
 A &= 600y - 2y^2 \\
 A' &= 600 - 4y \\
 600 - 4y &= 0 \\
 \frac{600}{4} &= \frac{4y}{4} \\
 \boxed{150} &= y
 \end{aligned}$$

The dimensions are
150 ft by 300 ft.

$$\begin{aligned}
 A &= (300 \text{ ft})(150 \text{ ft}) \\
 &= \underline{45\,000 \text{ ft}^2}
 \end{aligned}$$

A farmer wishes to create a rectangular field along a river with an area of 10,000 ft². What are the dimensions of the field that will require the least amount of fencing?

SOLUTION:



$$A = xy$$

$$\frac{10000}{x} = \frac{xy}{x}$$

$$y = \frac{10000}{x}$$

$$y = \frac{10,000}{141.42}$$

$$y = 70.7$$

$$P = x + 2y$$

$$P = x + 2\left(\frac{10,000}{x}\right)$$

$$P = x + \frac{20,000}{x}$$

$$P = x + 20,000x^{-1}$$

$$P' = 1 + 20,000(-x^{-2})$$

$$P' = 1 - 20,000x^{-2}$$

$$P' = 1 - \frac{20,000}{x^2}$$

$$1 - \frac{20,000}{x^2} = 0$$

$$1 = \frac{20,000}{x^2}$$

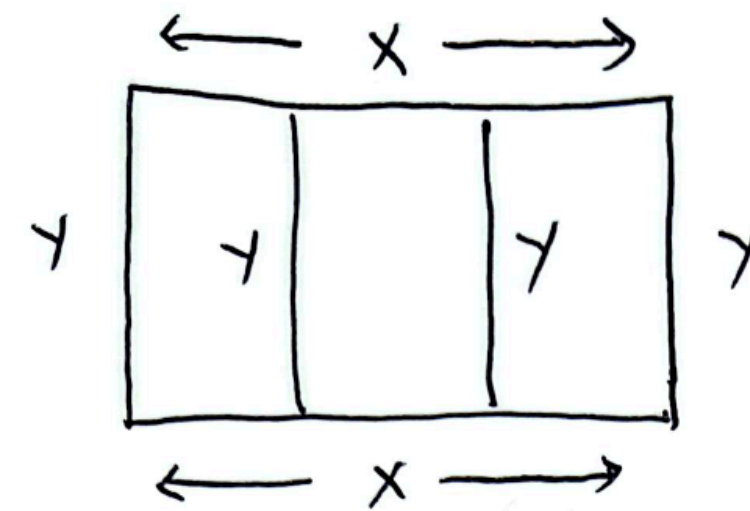
$$\sqrt{20,000} = \sqrt{x^2}$$

$$x = 141.42$$

The dimensions are 70.7 ft by 141.42 ft.

A farmer uses 1600ft of fencing to enclose a rectangular area which will be divided into 3 pens. What is the maximum total area of the 3 pens that he can enclose with the limited amount of fencing that he has available?

SOLUTION:



$$\begin{aligned}
 F &= 2x + 4y \\
 \frac{1600}{2} &= \frac{2x + 4y}{2} \\
 800 &= x + 2y \\
 800 - 2y &= x \\
 800 - 2(200) &= x \\
 800 - 400 &= x \\
 400 &= x
 \end{aligned}$$

$$\begin{aligned}
 A &= xy \\
 &= (800 - 2y)y \\
 A &= 800y - 2y^2 \\
 A' &= 800 - 4y \\
 800 - 4y &= 0 \\
 \frac{800}{4} &= \frac{4y}{4} \\
 200 &= y
 \end{aligned}$$

$$\begin{aligned}
 A &= xy \\
 A &= 400 \times 200 \\
 A &= 80,000 \text{ ft}^2
 \end{aligned}$$

The maximum total area is
80,000 ft².

**Thank you for
listening!**

